written answers for 5

5.6.1 Does one harmonic suffice or it is enough a single frequency variation?

It seems enough because the model already has a high R^2, and when adding the second harmonic (12-hour period), the R^2 of the model increases only slightly (exact numerical values shown in the code).

5.6.2 Does the plot show evidence for higher harmonics (12-hour component or more than a single frequency)?

After performing the residual diagnostics for the single-frequency model, plotting hour vs residual suggests that there may be a pattern. The one that stood out most to me was not the 12-hour component, but rather a 72-hour component. The FFT appeared to support this observation: the power associated with a 72-hour cycle was almost five times greater than that of the 12-hour cycle.

But we have to be careful when interpreting that result.

It looks like there may be a 72-hour cycle consisting of low peak, high peak, low peak. If you look at the residuals, there is a pattern every 72 hours: it starts low, rises, and then falls, and at hour 72 the pattern starts again. Basically, it resembles one complete oscillation over 72 hours, as if there were a “lower harmonic.” In other words, one might think that the fundamental frequency corresponds to 72 hours and the 24-hour component is a higher harmonic.

And the FFT showed that the 72-hour period has considerable power. It is the second strongest period in the spectrum, stronger than the 12-hour period.

However, the issue is that, to claim there is a true pattern, it must occur more than once. With only 3 cycles of data (72 hours = exactly 3 periods of 24 hours), it is very difficult to distinguish with statistical confidence between “there is a real slow variation with a period of about 72 hours” and “there is noise plus a small difference in amplitude between the three days, which my measurement tool (FFT) interprets as a long period because it does not have enough cycles to distinguish it more accurately.” We would need many more days of data for the FFT to reliably tell the difference.

5.8 What differences can you detect between the simple linear regression approach (e.3) and the FFT (e.7) results?

First, the similarities:

Both methods identified the same dominant period of 24 hours, and when comparing the amplitude that each method calculates for that frequency, the result matches exactly.

Now, the differences:

One difference is that linear regression only examines the frequencies that are chosen beforehand (in this case, 24 hours, and later I also tested with 12 hours), based on a prior hypothesis formed by looking at the original graph. FFT, on the other hand, examines all possible frequencies allowed by the sampling of the data, without requiring any hypothesis in advance. FFT detected a longer-period component (72 hours) that had not been considered when manually selecting frequencies for the regression.

Another difference is that each method produces a different type of result. Linear regression provides a complete and usable model, with the coefficients needed to predict temperature at any point in time. FFT provides an overview of which frequencies are present and their relative strength, which is useful for exploring and diagnosing the signal, but it is not by itself a ready-to-use predictive model.

They also answer subtly different questions. Linear regression tells you “how well does a specific frequency that I propose explain the temperature, and what are its exact amplitude and phase”. together with the R2. FFT, as I understood, tells you, “Out of all mathematically possible frequencies, which ones are actually present in my data, without me having to propose any of them beforehand?” These are different, although related, questions.